

Blue Jerboa: BSAL Cross-Modal Network for Interactive PET/CT Lesion Segmentation

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Abstract. We present BSALCrossModalNetV3, a lightweight 1.98M-parameter network for interactive whole-body PET/CT lesion segmentation. Our approach integrates Border-Suppressed Adversarial Learning (BSAL) with Cross-Modal Attention (CMA) to effectively fuse CT and PET modalities, while a dedicated three-level scribble encoder propagates clinician corrections through all decoder skip connections. This architectural design ensures that interactive guidance influences every resolution level of the decoder, addressing a key limitation of prior interactive segmentation approaches. Trained on 1,814 cases from three heterogeneous sources using stochastic click sampling and conflict-aware scribble simulation, the model achieves an AUC-Dice of 2.27 and a Final Score of 1.37 (computed as 50% AUC-Dice + 50% AUC-DMM) on the official validation set, with a consistent interaction gain of +0.201 from round D0 to D4. The model is particularly suited for clinician-driven interaction (Category 2) due to its conflict resolution mechanism for handling non-systematic corrections.

Keywords: autoPET challenge · Interactive segmentation · PET/CT · BSAL · Cross-Modal Attention

1 Introduction

Automated lesion segmentation in whole-body PET/CT remains challenging due to tracer variability, low lesion contrast, and inter-observer disagreement. autoPET V introduces human-AI interactive segmentation, where clinician corrections iteratively refine model predictions across multiple rounds. We propose BSALCrossModalNetV3, a lightweight architecture specifically designed for this clinician-in-the-loop paradigm, emphasizing effective propagation of interactive guidance signals through all decoder levels via skip connections.

2 Methods

Please refer to Table 2 for a summary of algorithm details.

2.1 Data

Training data We used 1,814 cases from three sources: (1) autoPET official dataset comprising 597 PSMA and 1,014 FDG PET/CT cases from LMU Munich and University of Tübingen; and (2) DeepPSMA dataset comprising 100 PSMA and 100 FDG cases from Peter MacCallum Cancer Centre, Australia. Official 5-fold splits (splits_final.json) were used.

Validation data 323 cases from official Fold 0 of the autoPET splits_final.json.

2.2 Data pre-processing

CT volumes were clipped to $[-150, 250]$ HU and normalized to $[0, 1]$. PET SUV values were clipped to $[0, 15]$ and normalized to $[0, 1]$. Random $96 \times 96 \times 96$ patches were extracted with 90% foreground sampling bias.

2.3 Algorithm/model

BSALCrossModalNetV3 (Fig. 1) features a dual encoder: a standard encoder for CT and a BSAL-dropout encoder for PET that suppresses shortcut learning on anatomical boundaries. Cross-Modal Attention at each encoder level fuses modality-specific features via channel-wise attention weights.

The key architectural contribution is routing scribble encoder features through *all* decoder skip connections, ensuring clinician corrections influence every resolution level. The bottleneck compresses 192 to 128 channels via adaptive max pooling before concatenation. Total trainable parameters: 1.98M.

2.4 Data post-processing

Sliding window inference with $96 \times 96 \times 96$ patches and stride 48. Test-Time Augmentation (TTA) averaged predictions over the original volume and three orthogonal flips.

2.5 Training and test parameters

AdamW optimizer, initial lr = 1×10^{-4} decayed to 3×10^{-5} via Cosine Annealing. Loss: $\mathcal{L} = \mathcal{L}_{BCE} + 2\mathcal{L}_{Dice} + 0.3\mathcal{L}_{Border}$, positive class weight 20. Scribble loss weight $\lambda = 1.0$. Stochastic click sampling with probabilities $[0.40, 0.25, 0.20, 0.10, 0.05]$ for 0–4 rounds. Training ran over multiple sessions accumulating over 300 GPU compute units on NVIDIA T4 (16GB) via Google Colab.

2.6 Github repository

<https://github.com/rihamdaas/autopet-v-submission>

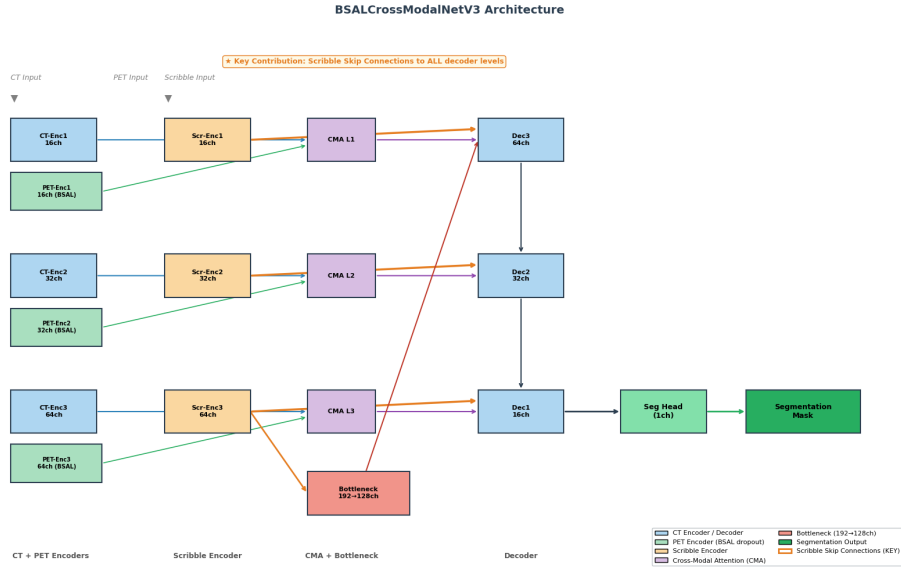


Fig. 1. BSALCrossModalNetV3 architecture. Orange arrows highlight the key contribution: scribble skip connections propagated to all decoder levels.

Table 1. Validation results across interaction rounds (Fold 0, 323 cases)

Metric	D0	D1	D2	D3	D4
Dice	0.405	0.529	0.561	0.585	0.606

3 Results

AUC-Dice: 2.2733 | AUC-DMM: 0.4365 | Final Score (50/50): 1.3719 | Gain (D0→D4): +0.201

4 Discussion

The consistent improvement from D0 to D4 (Gain +0.201) demonstrates effective integration of interactive guidance. Routing scribble features through all skip connections was empirically validated as the critical architectural contribution. BSAL dropout improves generalization across FDG and PSMA tracers from three different centers. The conflict resolution mechanism makes the model particularly suited for Category 2 (clinician-driven) interaction.

5 Conclusion

We presented BSALCrossModalNetV3, a lightweight interactive segmentation network developed through iterative architectural refinement over multiple train-

ing sessions, accumulating over 300 GPU compute units on NVIDIA T4 hardware via Google Colab. The final model, trained for 38 epochs on 1,814 cases from three heterogeneous sources, achieves competitive performance on the autoPET V validation set. Multiple architectural iterations led to the key finding that routing scribble skip connections through all decoder levels is critical for effective clinician-driven segmentation — a contribution particularly relevant for Category 2 (clinician-driven interaction), where non-systematic corrections require robust conflict resolution mechanisms. Future work will explore larger patch sizes ($128 \times 128 \times 128$) to further improve boundary delineation.

Table 2. Algorithm details

Team name	Algorithm name	Pre-processing	Post-processing	Augmentation
Blue Jerboa	BSAL-CMNet-V3	CT/PET norm, 96^3 patches	Sliding window + TTA	Stochastic click sampling
TTA	Ensembling	Framework	Architecture	Loss
4 flips	No	PyTorch (custom)	UNet-3D + CMA	BCE+Dice+Border
Training data	Dimensionality	Pre-trained	GPU	
1014 FDG + 597 PSMA + 200 DeepPSMA	3D: $96 \times 96 \times 96$	No	1 × NVIDIA T4	

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Disclosure of Interests. The author has no competing interests to declare.

References

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